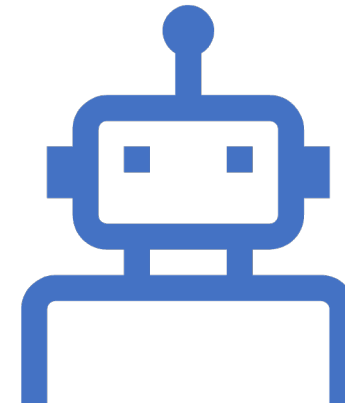




Emotion Analysis Web Science

Joemon M Jose



Britain's 'deep state' thwarted my plans, Liz Truss tells US hard-right summit

Former Conservative PM, whose tenure lasted 50 days, tells CPAC she fell victim to UK's 'establishment ... its bureaucrats and lawyers'



📷 Liz Truss with Nigel Farage at the CPAC summit on Wednesday. Photograph: José Luis Magaña/AP

Liz Truss, the former British prime minister, spoke at a far-right conference in America on Wednesday, styling herself as a populist who took on America's equivalent of the "deep state" in her own country.

Truss was among the headline speakers at this week's Conservative Political Action Conference at the National Harbor in Maryland. **CPAC** is billed as the biggest annual gathering of conservatives in the US but has in recent years embraced Donald Trump's brand of nativist-populism.

In Wednesday's opening session, an "international summit", the ex-PM sat side by side with **Nigel Farage**, former leader of the Brexit party, both with small Union flags on the table in front of them.

Other speakers included Steve Bannon, a former White House chief strategist associated with global far-right nationalist movements, and Richard Grenell, a former acting director of national intelligence under Trump. Officials from countries including Australia, Hungary and Japan also took part.

Why emotional analysis? Actionable insights!

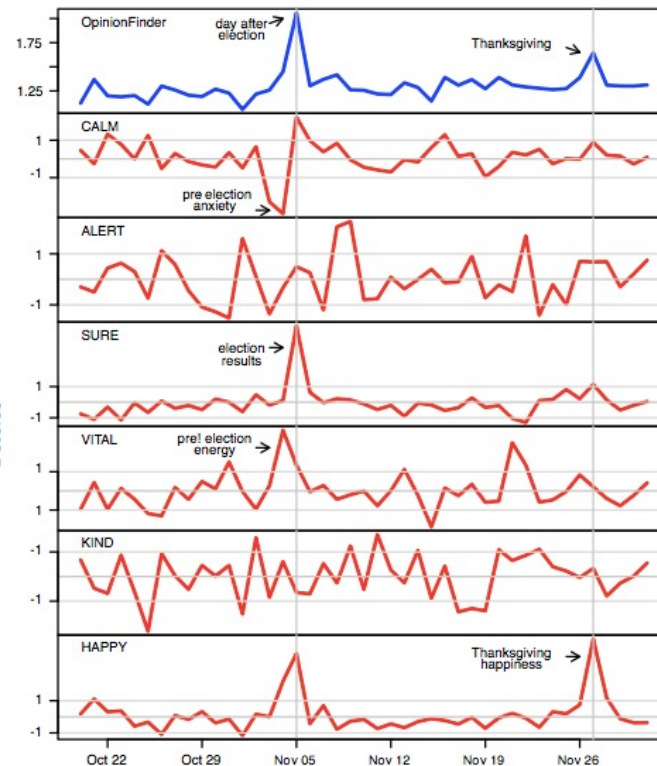
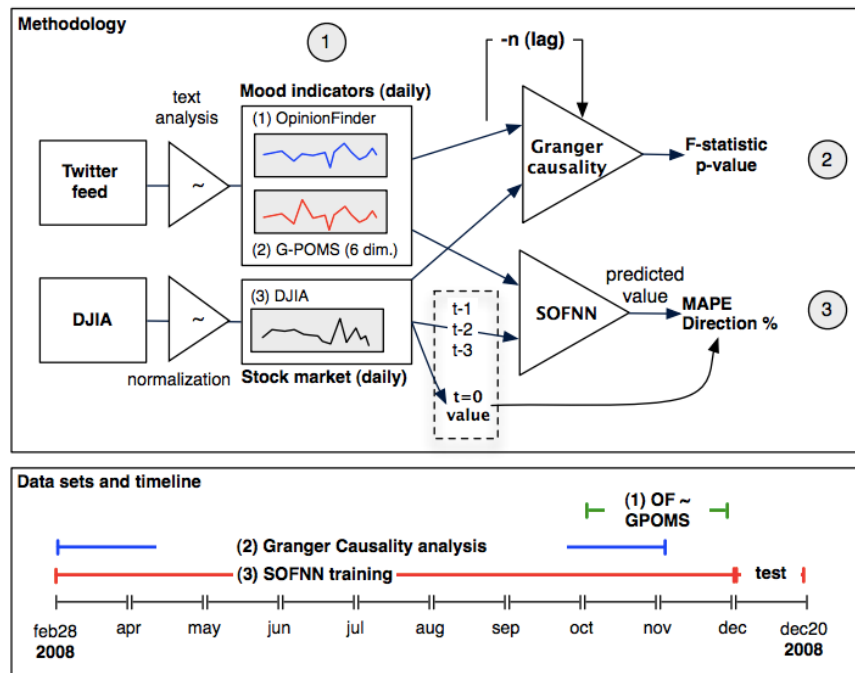
- Detecting
 - Attitude towards politicians, products, countries, ideas
 - Positive or negative polarity versus
 - Anger or excitement!

Other scenarios!

- Frustration of callers to a helpline
 - Positive or negative polarity versus
 - Frustration, sadness, anger vs happy
- Could we generate?
 - Emotion or moods for literacy tutors in children's story book domain
 - Emotion or moods for computer games
 - Personalities for dialogue systems to match the user

Twitter mood predicts stock market

Can societies experience mood states that affect their collective decision making?



Bollen, J et. Al, 2010

Sentiment analysis & variants

Actionable insights from text analysis!

sentiment analysis!

- “The House of Commons speaker is under increasing pressure after a debate on a ceasefire in Gaza descended into chaos.”
 - The sentiment of the sentence is negative,
 - as it indicates that the House of Commons speaker is facing increasing pressure due to a chaotic debate on a ceasefire in Gaza.



Variants

- Sentiment classification
 - whether a piece of text is positive, negative or neutral
 - Degree of intensity
 - $[-100, 100]$
 - $[-1, 1]$
 - Polarity – [positive, negative or neutral]

Opinion analysis

Determining from text, the speaker's opinion and target of the opinion

Opinion Mining provides granular information about the opinions related to words (such as the attributes of products or services) in the text.

N

The National  @ScotNational · 11h

...

 SNP MPs walk out of the House of **Commons** as it descends into chaos.

A huge row erupted after Speaker Lindsay Hoyle changed procedure to allow a Labour amendment to be debated, and then failed to come to the **Commons** to explain himself

Stance Analysis

- Author of text is in favour of, against of, or neutral towards a proposition or target
- For example, Brexit agreement
 - Is people supportive?
- **“The House of Commons speaker is under increasing pressure after a debate on a ceasefire in Gaza descended into chaos.”**
 - **If someone tweets**
 - Stand by the speaker amidst Gaza chaos.!
 - **In favour of speaker**
 - Speaker's failure worsens Gaza ceasefire chaos.
 - **Against of speaker**
 - Pressure mounts on speaker amid Gaza debate chaos.
 - Neutral towards speaker

Stance detection

- Is the task of automatically determining from text
 - whether the author of the text is in ***favour of, against of, or neutral***
 - **towards a proposition or target**
- Target
 - Person, organization, government policy, a movement, a product
 - E.g., infer from former Prime minister Boris Jonson's speeches that he is in favour of Brexit
 - E.g., analysing tweets identify people in favour of leadership change

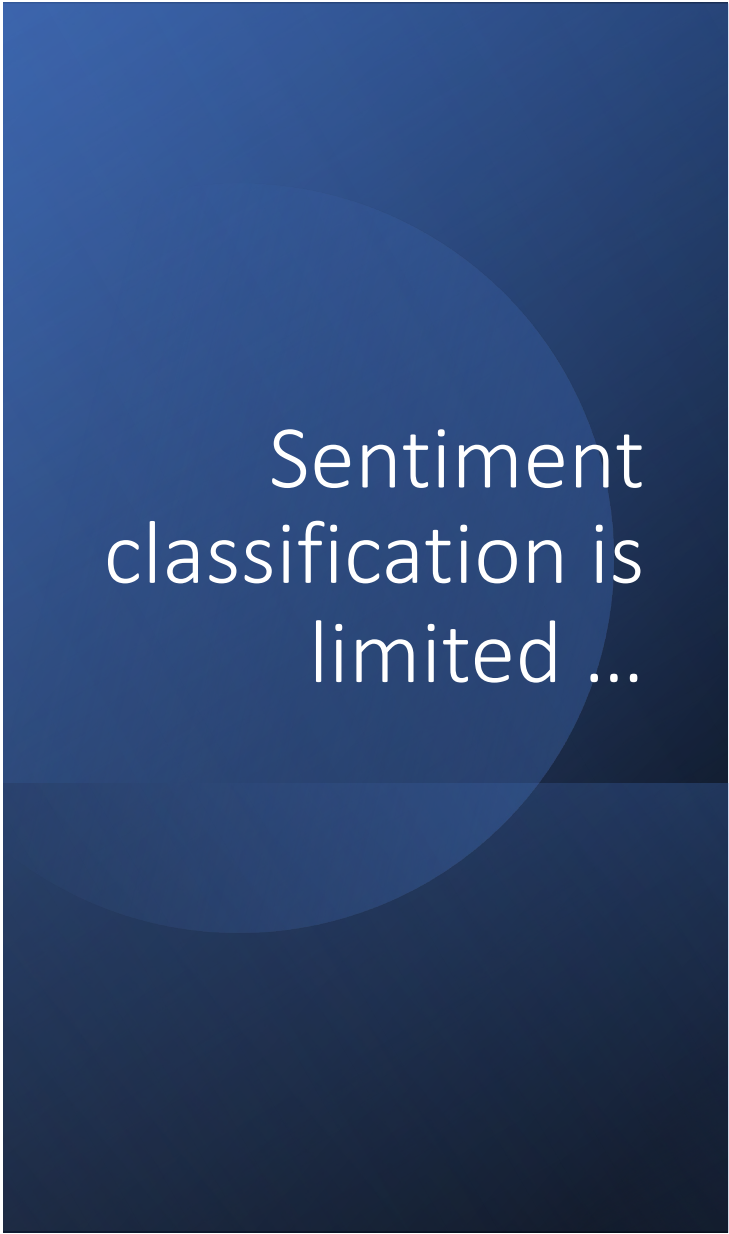
An active research area

Sentiment vs Stance

- Target:
 - Legalization of Abortion
- Tweet
 - The pregnant are more than walking incubators. They have rights too!
 - In favour of the target
 - Target – Pro-life movement
 - ??
- Target
 - Donald Trump
- Tweet
 - Donald Trump has some strengths and some weakness
 - neutral

Aspect Based Sentiment Analysis

- A sentence contains one or more entities,
 - each of which has a different polarity of emotion.
- For example, give a comment like
 - "Great food but the service is dreadful!",
 - the emotional polarity of entity "food" is "positive",
 - while the emotional polarity of entity "service" is "negative".
- Compared to sentence level sentiment analysis, ABSA can present
 - users with more precise and fine-grained sentiment information of entities
- You identify an aspect and the sentiment towards that aspect



Sentiment
classification is
limited ...

- Consider for instance:
 - Job stress / satisfaction
 - Creative expression in literature and music
 - Emotive topics such as domestic violence
 - factors in mental and cognitive health



Sentiment classification is limited ...

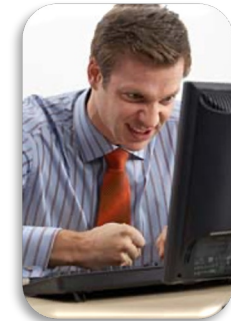
- Language serves social and interpersonal functions.
 - Affective meaning is key for human interaction and a prominent characteristic of language use.
- This extends beyond
 - *opinions vs. factual or polarity distinctions*
 - into multiple phenomena:
 - **emotion, mood, personality, attitude, credibility, volition, veracity, friendliness, etc.**
- **Emotion:**
 - *angry, sad, joyful, fearful, ...*
- **Emotion**
 - **What are the emotion expressed in the text?**
- However, recognition, characterization, or generation of affect states,
 - involves analysis of affect-related conditions, experiences, and activities.

What are emotions?

- *“episodes of interrelated, synchronized changes in the states of all or most of the five organismic subsystems*
 - Information Processing, support, executive, action, monitor
- *in response to the evaluation of an external or internal stimulus event as relevant to major concerns of the organism”**

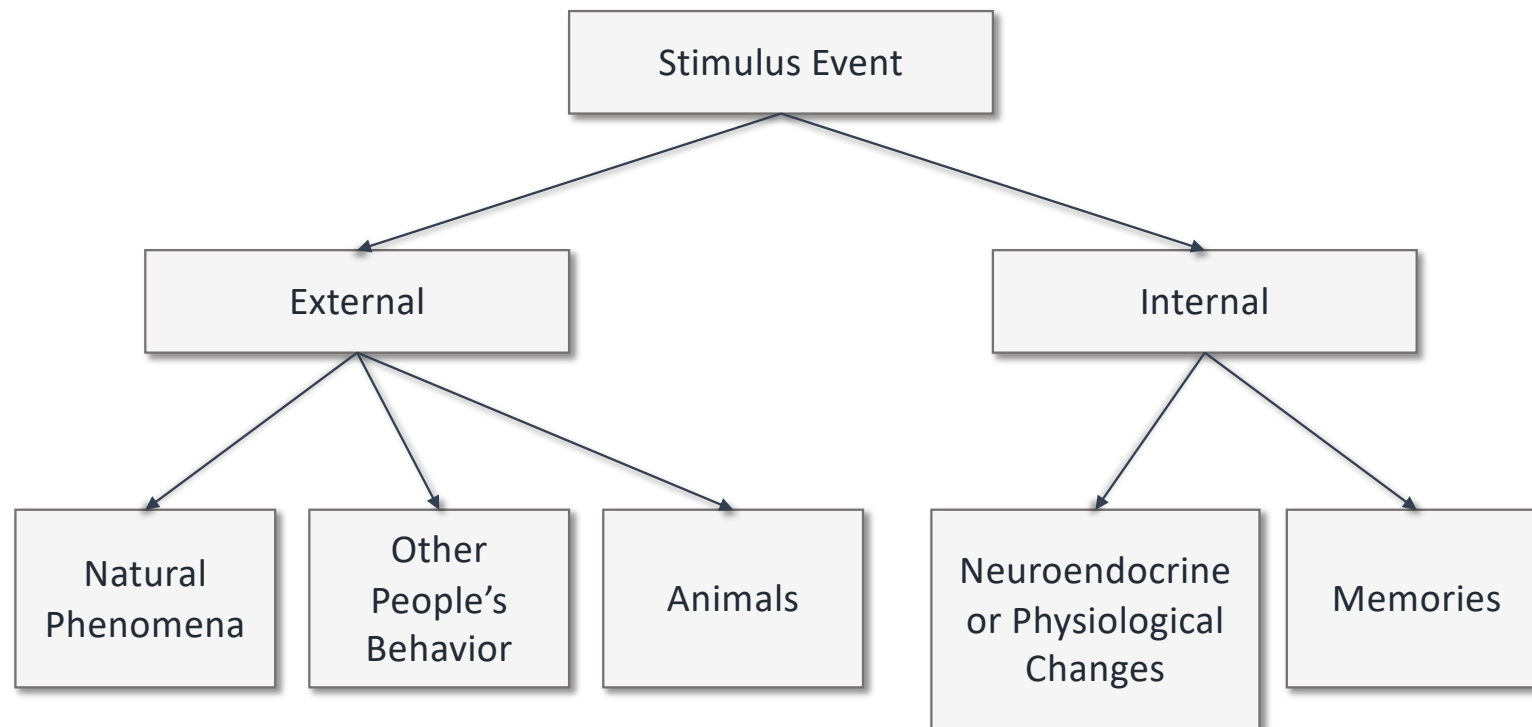
emotional behaviours and experiences can be used
for exploring Human-Centred problems such as

healthcare
educational
political or
artistic domains



Scherer, K. R., 2005. What are emotions? And how can they be measured? In Social Science Information. 44, 4, 695-729.

What triggers emotions?



Textual emotion

- Researchers in psychological science believe that
 - individuals have internal mechanisms for a limited collection of responses, usually
 - happy, sad, anger, disgust, and fear
- Our aim is to analyse the text and detect the emotion
 - expressed by the author or
 - the emotion potentially felt by the reader
- Linguistic sensing of affective states can be used for
 - Understanding social issues expressed through social media

Example

- “Really happy with this purchase”
 - express the positive emotions from a customer about the purchase of product.
 - The term “Really” intensifies the emotion expressed by the customer.
 - In this case it implies it is more positive emotions.
- “Really disappointed. Alexa has to be plug-in to wall socket all the time. My fault is for not checking”.
 - Negative emotion

Emotion paradox

- People believe that they know an emotion when they see it, and
 - assume that emotions are discrete events that can be recognized with some degree of accuracy,
 - but scientists have yet to produce a set of clear and consistent criteria for indicating
 - when an emotion is present and when it is not.
 - one solution to this paradox:
 - People experience an emotion when they conceptualize an instance of affective feeling.
 - In this view, the experience of emotion is an act of categorization,
 - guided by embodied knowledge about emotion.
 - This model of emotion experience that has much in common with the social psychological literature
 - on person perception
-
- **Lisa Feldman Barrett**, Personality and Social Psychology Review, 2006, Vol. 10, No. 1, 20–46

Categorical approach

- This approach of emotion involves placing emotions into categories or
 - into distinct classes that are basic and universally recognized
- The emotions are independent and also
 - depend on how someone experience / perceive a situation
- Thus, emotions can be categorized into six basic fundamental dimensions i.e.,
 - *happiness, surprise, fear, sadness, disgust and anger*
 - by Paul Ekman
 - Known as universal emotions

Ekman Universal emotion categories

- There are 6 emotion categories that are widely used to describe humans' basic emotions, based on facial expression:
 - anger, disgust, fear, **happiness**, sadness and **surprise**.
- Categorical theories
 - Emotions are discretely and differently constructed and
 - all humans are thought to have an innate set of basic emotions
 - that are cross-culturally recognisable

OCC model

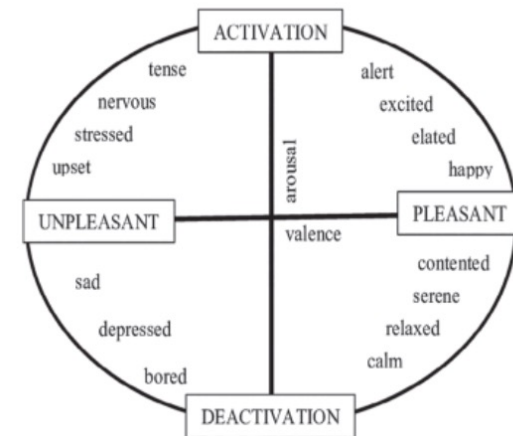
- Emotions have been categorised into 22,
 - *6 universal Paul Ekman emotions*
 - +
 - 16 additional emotions were
 - *envy, relief, appreciation, self-reproach, shame, reproach, pity, admiration, disappointment, grief, gratification, fears-confirmed, gloating, hope, like and dislike.*

Criticism

- In the categorical approach, emotional states are restricted to a limited number of distinct types and
 - it can be difficult to resolve a complex emotional situation or mixed emotions.
- Appraisal theory describes how different emotions,
 - in various participants and on different times,
 - can arise from the same event
- Essentially, our appraisal of a situation causes an emotional, or affective, response that is going to be based on that appraisal.

Dimensional approach – Russell's circumplex model

- it considers emotional states to be bound to each other
 - rather than to be independent.
- Russell's circumplex model represent the emotions in two-dimensional model.
 - As emotions are not independent it distinguished as Arousal
 - Activation and Deactivation and
 - Valence
 - Pleasantness and Unpleasantness
 - where the dimension Arousal refers to how excited or apathetic an emotion is and
 - the dimension Valence refer to how positive and negative emotion



Sentiment vs emotion

- Considered to be formed and retained for larger period of time
- Emotion brief episodes and shorter in length
- Sentiments are target centred
 - Hence directed
- Emotions are not target centred
- Multiple emotions are possible for a text
 - Only single sentiment
 - Positive, negative or neutral

So far

- We discussed what is emotion?
- How is it represented:
 - Categorical
 - 6 basic and universal emotions, Paul Ekman
 - OCC model – 22 emotions
 - Dimensionality approach – Circumplex model
- **For the rest of our discussion,**
 - **we will focus on categorical representations**

Emotion detection

- Aim is to identify users' emotional states from a short text by classifying emotions into one of the finite categories
 - Anger, fear, happy, sad, disgust, and surprise
- Emotion distribution can also be learned!
 - For a given text, a mixture of emotions could also be assigned
 - 'guy is a crazy guy', - disgust 50%, surprise 50%
 - '7 dead in apartment building fire'
 - Anger 10%, disgust 1%, fear 30%, joy 0, sadness 50% surprise 5%
 - 'Teacher charged with sexual assault'
 - Anger 20%, disgust 45%, fear 10%, joy 0%, sadness 15% surprise 10%

Text emotion detection

motivation

- because of the naturally vague and ambiguous human language is,
 - the emotion detection can be highly "context-sensitive and complex"
- Emotion analysis is a convoluted task, even for human beings,
 - The task will be much easier when emotion is expressed explicitly, but
 - in reality, the majority of texts are subtle, ambiguous, and
 - some words have more than one meaning, and
 - more than one word expresses the same emotions, and, in addition,
 - some emotions can exist simultaneously
 - due to the various cultures, gender, and context of people who authored the texts.

Applications relevant to our discussion so far!

- Given an event, for example, the chaos at the house of commons yesterday
 - People reactions and in general the emotion distribution
- Budget reports and other discussions
 - People reactions and the economic mood
- Given the comments on stocks, can we predict the price movements!
- By analysing the horse racing comments,
 - the model can learn about the information of winning horse and
 - would be able to give a reasonable prediction based on the emotion.



emotion classification

- In the case of sentiment analysis, this task can be tackled using
 - lexicon-based methods, machine learning, or a rule-based approach
- In emotion recognition task, the most common approaches are
 - Keyword-based detection
 - Seed opinion words and find synonyms & antonyms in WordNet
 - WordNet is a lexical database of semantic relations between words that links words into semantic relations including synonyms, hyponyms, and meronyms
 - Learning based detection/Machine Learning

Lexicon

- Lexicons are linguistic tools for automated analysis of text
- Most common
 - Simple list of terms associated to a certain class of interest
 - Terms can be weighted according to their strength of association with a given class
 - Classification by counting

	word	emotion	association
0	aback	anger	0
1	aback	anticipation	0
2	aback	disgust	0
3	aback	fear	0
4	aback	joy	0
5	aback	negative	0
6	aback	positive	0
7	aback	sadness	0
8	aback	surprise	0
9	aback	trust	0
10	abacus	anger	0
11	abacus	anticipation	0

EmoLex

- Each word and association with an emotion
- Aback has zero association with anger

NRC hashtag Lexicon

		Npos	Nneg
• .#great	8.988	6213	0
• .#classy	8.841	5364	0
• .#wonderful	8.735	4823	0
• .#perfection	8.607	4246	0
• .#perfect	8.521	3895	0
• .#excellent	8.385	3400	0
• .#positive	8.351	3285	0
• .#heavenly	8.319	3183	0
• .#fabulous	8.102	2561	0
• .#amazing	8.024	2368	0
• .#ideal	7.986	2280	0
• .#nice	7.984	2276	0
• !#amazing	7.911	2115	0
• !#positive	7.810	1912	0
• .#superb	7.663	1651	0
• !#ideal	7.603	1554	0
• !#perfect	7.568	1501	0
• ... #great	7.539	1458	0

- <score> is a real-valued sentiment score:
score = $\text{PMI}(w, \text{pos}) - \text{PMI}(w, \text{neg})$, where
- PMI stands for Pointwise Mutual Information between a term w and the positive/negative class;
- <Npos> is the number of times the term appears in the positive class, i.e., in tweets with positive hashtags;
- <Nneg> is the number of times the term appears in the negative class, i.e., in tweets with negative hashtags.

$$\text{pmi}(x; y) \equiv \log_2 \frac{p(x, y)}{p(x)p(y)} = \log_2 \frac{p(x|y)}{p(x)} = \log_2 \frac{p(y|x)}{p(y)}$$

What is pmi?

- Pointwise mutual information
 - The PMI of a pair of outcomes x and y
 - belonging to discrete random variables X and Y
 - quantifies the discrepancy between the probability of their coincidence
 - given their joint distribution and
 - their individual distributions, assuming independence.

Lexicon based approaches?

- ***“I love horror books”***
- $f(\text{love}) \times 1.0 + f(\text{horror}) \times 0.3 + f(\text{books}) \times 0.5 = 1.8$ for positive
- $f(\text{love}) \times 0.0 + f(\text{horror}) \times 0.7 + f(\text{books}) \times 0.5 = 1.2$ for negative
- Decision function
 - Classify one with maximum value
- Transparency!
 - Each prediction can be explained
 - Analysing terms that were present in the text

term	positive	negative
love	1.0	0.0
horror	0.3	0.7
books	0.5	0.5

Pros & Cons

- **Limited Vocabulary Coverage:**

- Lexicon-based approaches rely on predefined emotion lexicons,
- which may not cover the entire spectrum of human emotions or
- may not include culturally specific expressions of emotion.
- This can lead to difficulties in accurately detecting and categorizing emotions, especially those outside the lexicon.

- **Lack of Context Sensitivity:**

- These approaches often do not take into account the context in which words are used,
- leading to misinterpretation of emotion.
- For example, the word "kill" might be associated with negative emotions in general, but in certain contexts (e.g., "I killed it in my presentation"), it might indicate success or accomplishment rather than negativity.

- **Difficulty with Ambiguity and Polysemy:**

- Words can have multiple meanings and can be associated with different emotions depending on the context.
- Lexicon-based approaches may struggle to disambiguate such cases accurately.

So far

- We look into the definitions
- We looked into a lexicon-based approach

Datasets for evaluation

First issue to consider is what emotional representation to use!

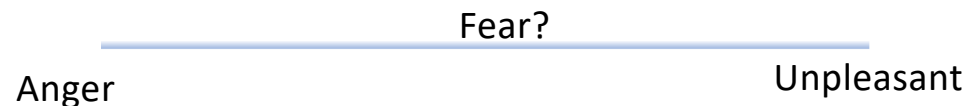
Basic Emotion Categories

- **Basic (primary) emotions**
 - `Pure' emotional states
 - E.g. Ekman's Big 6:
 - Fear, anger, **happiness**, sadness, **surprise**, disgust
- considered from text
 - Only one positive class – happiness
 - What about surprise?
- When analyzing text?

classes

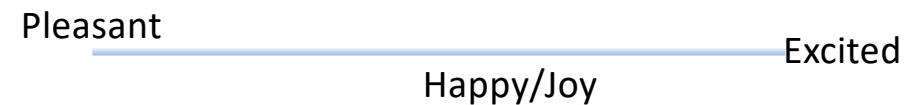
-ve

- **Anger**
- **Fear**/anxiety
- uninterested/nervous/**sadness**/
unpleasant
- How do we justify this?
 - Degree of negativity

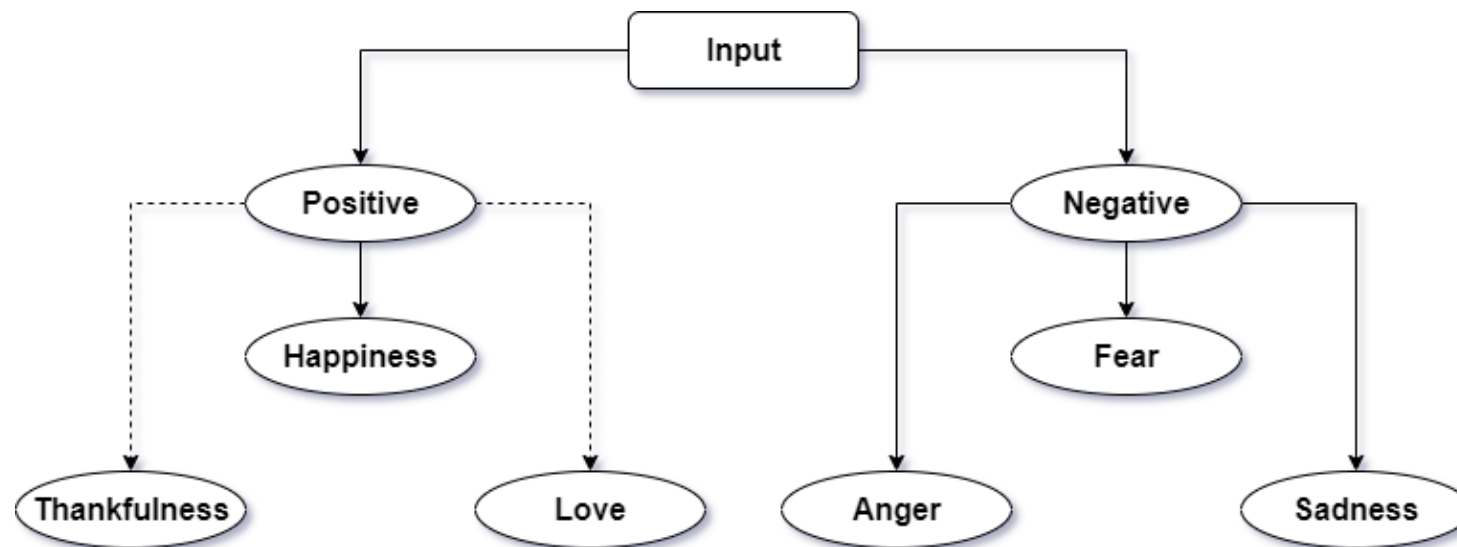


+ve

- Excited
- **Happy**/love/joy
- Surprise/Pleasant/Interest/grate
ful/**Thankful**
- How could we justify this?



Hierarchy of emotions



How do we develop models

- Capture ground truth
 - Data with corresponding annotations
 - We need quite large sets
- Manually doing is difficult!
 - Why?
- Automatic methods
 - Labels from users
 - Similar to distant supervision we used!!

Data Collection

- Label ambiguity
 - uncertainty,
 - incompleteness
 - mistakes among the ground truth label due to annotators expertise or task's difficulty
- Cross-label interference
 - Happy vs surprise/pleasant
 - "Wow, I can't believe it! I just found out I won the lottery! This is the happiest day of my life!"
 - Anger vs fear
 - "Every time you lie to me, I can feel the rage boiling inside me, but beneath it all, there's this gnawing fear that I'll never be able to trust you again."
 - Fear vs unpleasantness
 - "As the eerie shadows crept closer in the dimly lit alley, a shiver ran down my spine, filling me with a sense of unease and dread."

datasets

- Data collection objectives
 - Why do we collect data?
 - What kind of data?
 - How to annotate?

Newsworthiness model

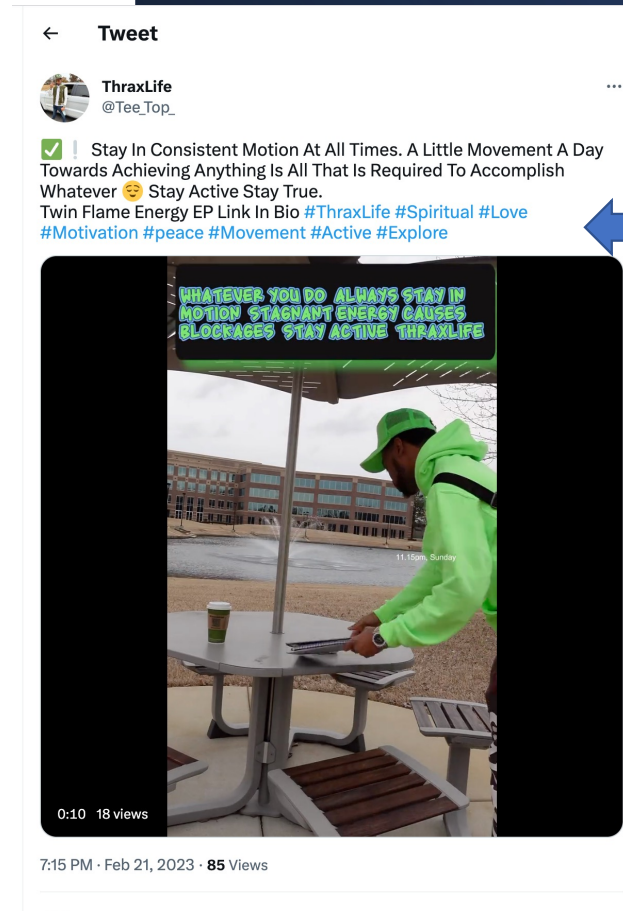
- We have used distant supervision
- Can we exploit such techniques for creating a dataset for emotion analysis?

Tweets

-  Stay In Consistent Motion At All Times. A Little Movement A Day Towards Achieving Anything Is All That Is Required To Accomplish Whatever 😊 Stay Active Stay True.

Twin Flame Energy EP Link In Bio #ThraxLife
#Spiritual #Love #Motivation #peace #Movement
#Active #Explore <https://t.co/JCagFG8Wh1>

- ['#Active', '#Explore', '#Love', '#Motivation', '#Movement', '#Spiritual', '#ThraxLife', '#peace', 'accomplish', 'achieving', 'active', 'anything', 'bio', 'consistent', 'day', 'energy', 'flame', 'link', 'little', 'motion', 'movement', 'required', 'stay', 'times', 'towards', 'true', 'twin', 'whatever']



another

- I attract into my life what I am, not what I want. I am healthy, I am happy, I am loved, I am abundant. Be the energy of the life you desire. #love



Another two!

-
- 🥰ENJOY MY FRIEND🥰
- #MaykiGTI #Eminem #DrDre #SnoopDogg #artist #rap #music ##موسيقى #מסיקת Love #Bitcoin #ETH #Web3
<https://t.co/a1INwElceU>
- Enjoy this #love & happiness life #quoteoftheday, & pass it on! <https://t.co/DeCytjhYWd>
- 📱Phone Case for Cats lovers
- 📱Check our new designs
- 📱Link: <https://t.co/B6z2p89QcK>
-
- #iphonecase #iphone #phonecases #smartphone #iphoneaccessories #iphonecamera #iphoneshot
#smartphoneaccessories
- #love #fashion #art #photography #happy
- 📱Phone Case for Cats lovers
- 📱Check our new designs
- 📱Link: <https://t.co/B6z2p89QcK>

Fear!

- Should turbulence be something to be legitimately scared of?
<https://t.co/3wH89gsvrZ>
#turbulence #fear #flying
<https://t.co/W0LRwnoQgm>

← Tweet



Capt Tom Bunn LCSW
@flightwit

...

Should turbulence be something to be legitimately scared of?
goo.gl/3mQatM #travel #turbulence #fear #flying



4:31 PM · Nov 2, 2016



fear



- Your coordinates are advancing too quickly for #fear to be part of the equation.
- #US #Stocks #Bearish #Fear-#Fed More #Hawkish #Geopolitical #War Shadow
- #WallStreet #Bears Strike cont.>June/22 Lows
- #USD-#Gold
- Fut
- #Nasdaq>12.3K(12.114)
- #SP500<4.05K(4010
- #DJIA<33.5K(33.193)
-
- #DXY>104(104.10)
- #Gold>1.83K(1835)
-
- ATA
- #Trends #Patterns 1st <https://t.co/hWHA0Ufhga>
-

emojies

- @catherinekelley @RheaRipley_WWE You're a nice person, Cathy... so imma be happy for you...   
<https://t.co/bgC82decqi>

- The blue heart emoji  is commonly used to express trust, harmony, peace, and loyalty.

- Well, I can honestly say this evening has been joyous. Had happy thoughts and many of them, really nice chat with friends on WhatsApp, a huge smile, yeah tonight's been good  



Emotion oriented hashtags?

- So far, we have seen
 - Text with hashtags and emojis
- Could we exploit such hashtags?
 - ?
- If we were to use such hashtags, what rule(s) to use?
 - ?

Hashtag based data collection

This approach yields a completely automatic scheme for labelling a massive repository of Twitter messages

example

- Emotions are complex concepts with fuzzy boundaries with individual variations in expression and perception



SOFIE @lionessofie1 · 6h

i had a good day today and i feel happy x

get me x @justinbieber ❤️🌙 #getme #justinbieber #newmusic #cover
#waves #fender #beliebers #changes #happy #excited #lit #blue #sing
#play #grateful #boston #canada #chill #dream

Tweets as annotated data

- Assumption
 - “emotional hashtags inserted by the author indicate the main emotion expressed by their Twitter message”



Sherry Dickinson @SherryDickins17 · Feb 11

Happy [#Hugday](#)...I wish this day will come back in your life again and again.[#TakeCare](#) of your favourite people with [#Smartness](#) and be [#Happy](#) all time.Wait for [#KissDay](#) ,this day is [#Comming](#) afer some hours.[#Love](#) you all.

Conflicting hashtags

- What do we do in case conflicting hashtags?



Neelita @neeldreamz · Apr 1, 2015

The first step towards Dream... #FirstWorkingDay #Nervous #Excited
#Happy #Determined



What to annotate?

Data Labelling Process

Identify emotion hashtags

Twitter API

Collect labelled Tweets

Data processing

Labelled Data

- Identify hashtags
- first need to define a list of emotion hashtags to collect emotion-labeled messages
 - 28 affect words from Circumplex model
- collect tweets which contain one or more hashtags that fall in the defined list of emotions hashatgs



ContemptraINN 🌹
@CONTEMPRA_INN

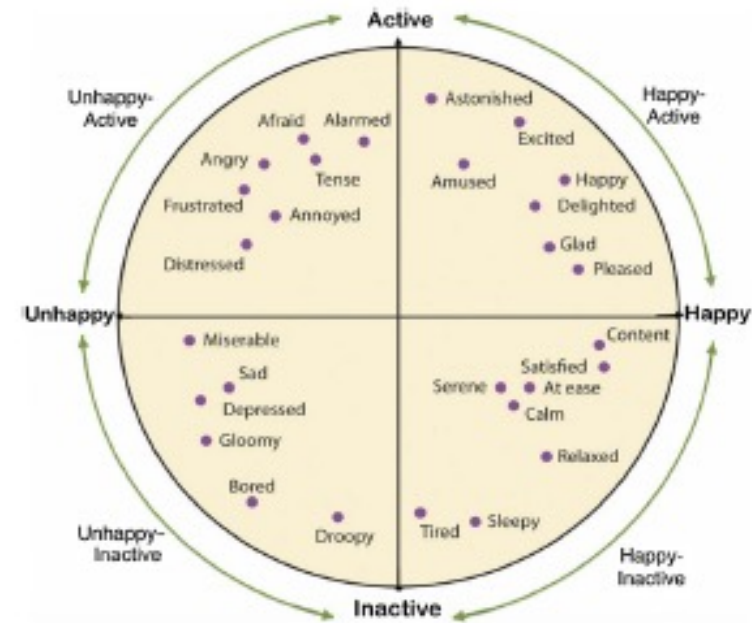
May you find **#Happy #Surprises** in Life 🌈 via
@KShostakovskaya @CONTEMPRA_INN 🌹

```
lf.fl_pleasant = ['positive', 'pleasant', 'good', 'nice', 'glad', 'relaxed', 'serene', 'comfort', 'relax', 'satisfied', 'enjoyable', 'delightful', 'wonderful', 'amaze', 'admiration', 'pleased', 'satisfied', 'tease', 'relaxed']
lf.fl_excitement = ['excited', 'excitement', 'exciting', 'energizing', 'galvinizing', 'thrill', 'passion', 'astonished', 'exhilaration', 'inspiring', 'moving', 'sensational', 'ecstasy', 'pride', 'euphoria', 'zeal', 'zest', 'enthusiasm']
lf.fl_happy = ['happy', 'joy', 'love', 'happiness', 'loving', 'laughing', 'delighted', 'bliss', 'fondness', 'jubilation']
lf.fl_sad = ['concerned', 'pathetic', 'cry', 'negative', 'sad', 'shy', 'frustration', 'sadness', 'sorry', 'surprise', 'frustrated', 'bad', 'bored', 'afraid', 'tense', 'anxious', 'gloomy', 'annoyed', 'offended', 'tired', 'tear', 'miserable', 'disap']
lf.fl_angry = ['fury', 'frighten', 'terror', 'anger', 'rage', 'outrage', 'angry', 'enraged', 'mad', 'ballistic', 'piss off', 'horror', 'alarmed', 'terrified', 'horror', 'panic', 'tragedy', 'disgusted', 'loathing', 'aggression']
lf.fl_fear = ['fear', 'disgust', 'depression', 'depressed', 'distress', 'contempt', 'anxious', 'tearful', 'panic', 'obscene', 'detestable', 'afraid', 'frustrated', 'annoyed']
lf.fl = self.fl_pleasant + self.fl_excitement + self.fl_happy + self.fl_sad + self.fl_angry + self.fl_fear
```

Expanding seed words

Table1: Parrot's Emotion Taxonomy

Primary emotions	Secondary emotions	Tertiary emotions
Love	Affection	Caring, sentimentality, adoration, fondness, compassion, tenderness, love, attraction, affection, liking.
	Lust	Desire, arousal, passion, Infatuation, lust.
Joy	Cheerfulness	Euphoria, satisfaction, gladness, ,bliss, happiness, Jubilation, delight, elation, enjoyment, joviality, joy, Amusement ecstasy, jolliness, gaiety, glee, cheerfulness.
	Zest	Thrill, excitement, enthusiasm, exhilaration. zeal, zest.
	Contentment	Pleasure, Contentment.
	Pride	Triumph, Pride.
	Optimism	Optimism, hope, Eagerness.
Surprise	Surprise	Astonishment, surprise, amazement.
	Irritation	Agitation, annoyance, irritation, grouchiness aggravation,
Sadness	Sadness	Gloom, grief, melancholy despair, glumness, misery, unhappiness, hopelessness, sadness, depression, sorrow, woe,
	Disappointment	Dismay, displeasure , disappointment
	Shame	Regret, shame, remorse, guilt
Anger	Exasperation	Frustration, exasperation,
	Rage	Bitterness, loathing, wrath, dislike, hostility, spite, resentment rage, ferocity, hate, scorn, fury, anger, vengefulness, outrage.
	Disgust	Contempt, revulsion, disgust.
	Suffering	Hurt, agony, anguish suffering,



```
lf.fl_pleasant = ['positive', 'pleasant', 'good', 'nice', 'glad', 'relaxed', 'serene', 'comfort', 'relax', 'satisfied', 'enjoyable', 'delightful', 'wonderful', 'amaze', 'admiartion', 'pleased', 'satisfied', 'atease', 'relaxed']
lf.fl_excitement = ['excited', 'excitement', 'exciting', 'energizing', 'galvinizing', 'thrill', 'passion', 'astonished', 'exhilaration', 'inspiring', 'moving', 'sensational', 'ecstasy', 'pride', 'eupharia', 'zeal', 'zest', 'enthusiasm']
lf.fl_happy = ['happy', 'joy', 'love', 'happiness', 'loving', 'laughing', 'delighted', 'bliss', 'fondness', 'jubilation']
lf.fl_sad = ['concerned', 'pathetic', 'cry', 'negative', 'sad', 'shy', 'frustration', 'sadness', 'sorry', 'surprise', 'frustrated', 'bad', 'bored', 'afraid', 'tense', 'anxious', 'gloomy', 'annoyed', 'offended', 'tired', 'tear', 'miserable', 'disap']
lf.fl_angry = ['furry', 'frighten', 'terror', 'anger', 'rage', 'outrage', 'angry', 'enraged', 'mad', 'ballistic', 'pissoff', 'horror', 'alarmed', 'terrified', 'horror', 'panic', 'tragedy', 'disgusted', 'loathing', 'aggression']
lf.fl_fear = ['fear', 'disgust', 'depression', 'depressed', 'distress', 'contempt', 'anxious', 'tearful', 'panic', 'obscene', 'detestable', 'afraid', 'frustrated', 'annoyed']
lf.fl = self.fl_pleasant + self.fl_excitement + self.fl_happy + self.fl_sad + self.fl_angry + self.fl_fear
```

How could we collect seed words?

- Which api to use?
 - Twitter's REST API:
 - Allows you to search terms or get tweets based on specific parameters.
 - This doesn't give you live data. More useful if you want to do some analytics on historical data.
 - Twitter's Streaming API:
 - Gives you live tweets data as per your request and keeps sending it until you ask it to stop.

Data Processing & Rules

Some scenarios to consider



ContemptraINN 🌹
@CONTEMPRA_INN

May you find **#Happy #Surprises** in Life 🌈 via
@KShostakovskaya @CONTEMPRA_INN 🌹

- **Consider tweets only with hashtags at the end**
 - If it is actually interleaved,
 - then probably actual part of the content
- Give direct access to author's intent

#iphonecase #iphone #phonecases #smartphone #iphoneaccessories #iphonecamera #iphones
#love #fashion #art #photography #happy

🔗 Phone Case for Cats lovers

🔗 Check our new designs

🔗 Link: <https://t.co/B6z2p89QcK>



toskeyy @toskeyy_ · 6h

An idea a friend gave me (Catnap was too curious) **#miau #cat #love**
#catlover #happy #life #meow #smile #happiness #sweet #style #catnap
#dogday #miauu (I didn't finish painting it well, but it's something)

so much?



something?

My friend Luna needs a



How nice your
gesture!

Rule 1



- If the text is annotated with a hashtag-
 - For an emotion dimension
- Give direct access to author's intent
- Example,
 - I attract into my life what I am, not what I want. I am healthy, I am happy, I am loved, I am abundant. Be the energy of the life you desire. #love
 - Assign the category love
- And also
 - Any hashtag for that seed list
 - Assign it to list

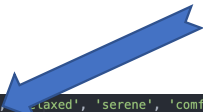
Rule 1 extended

In fact, we can also use any of the hash-tagged seed terms –
they all fall into same dimension

For the class **Excite**

#excited, #energizing,

Through this, we will get diversity of expressions which are semantically closer



```
lf.fl_pleasant = ['positive', 'pleasant', 'good', 'nice', 'glad', 'relaxed', 'serene', 'comfort', 'relax', 'satisfied', 'enjoyable', 'delightful', 'wonderful', 'amaze', 'admiration', 'pleased', 'satisfied', 'tease', 'relaxed']
lf.fl_excitement = ['excited', 'excitement', 'exciting', 'energizing', 'galvinizing', 'thrill', 'passion', 'astonished', 'exhilaration', 'inspiring', 'moving', 'sensational', 'ecstasy', 'pride', 'euphoria', 'zeal', 'zest', 'enthusiasm']
lf.fl_happy = ['happy', 'joy', 'love', 'happiness', 'loving', 'laughing', 'delighted', 'bliss', 'fondness', 'jubilation']
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lf.fl_angry = ['fury', 'frighten', 'terror', 'anger', 'rage', 'outrage', 'angry', 'enraged', 'mad', 'ballistic', 'piss off', 'horror', 'alarmed', 'terrified', 'horror', 'panic', 'tragedy', 'disgusted', 'loathing', 'aggression']
lf.fl_fear = ['fear', 'disgust', 'depression', 'depressed', 'distress', 'contempt', 'anxious', 'tearful', 'panic', 'obscene', 'detestable', 'afraid', 'frustrated', 'annoyed']
lf.fl = lf.fl_pleasant + lf.fl_excitement + lf.fl_happy + lf.fl_sad + lf.fl_angry + lf.fl_fear
```

just seed words

```
lf.fl_pleasant = ['positive', 'pleasant', 'good', 'nice', 'glad', 'relaxed', 'serene', 'comfort', 'relax', 'satisfied', 'enjoyable', 'delightful', 'wonderful', 'amaze', 'admiration', 'pleased', 'satisfied', 'tease', 'relaxed']
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lf.fl_angry = ['fury', 'frighten', 'terror', 'anger', 'rage', 'outrage', 'angry', 'enraged', 'mad', 'ballistic', 'pissed off', 'horror', 'alarmed', 'terrified', 'horror', 'panic', 'tragedy', 'disgusted', 'loathing', 'aggression']
lf.fl_fear = ['fear', 'disgust', 'depression', 'depressed', 'distress', 'contempt', 'anxious', 'tearful', 'panic', 'obscene', 'detestable', 'afraid', 'frustrated', 'annoyed']
lf.fl = self.fl_pleasant + self.fl_excitement + self.fl_happy + self.fl_sad + self.fl_angry + self.fl_fear
```

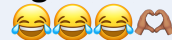
💋💋 Live a good life meet slowly. 💖💖💖💖💖

Emojies!

@muhcaw Auughh I wanna like the game so bad, but the dialogue is PAINFUL, and it crashed twice in about an hour last time I played it 😭

cry

I can't wait for my friend to come I be so bored and we be in our own lil world laughing at everything



funny

@anyabeckVA Aaah! Those are both so wonderful! Bass guitar is cool as heck and knitting is something I've always admired. You can make so many cute things 😊

emoticons

<https://emojipedia.org/twitter/>



Emotion	Emoticons
sadness	>:[:-{ :(-c:c :-< :-{ :[:{
anger	:- :@> :{
joy	:) ;) =) :] :P :-P ;P :D ;D :> :3 :-} ;-) :^) :o) ;^) :-D :->
surprise	:-o :-O o_O O_o :\$
disgust	D:< D: D8 D; D= DX v.v

emoticons

- Can also be exploited!
- Similar emoticons grouped together
- Rule 2
 - Annotate using emoticons, if it is at the end

```
self.emoticonsAngry = [':@', '>:(', '>:[', ':/', ':-/', '=/', ':S']
self.emoticonsFear = ["D-'", 'D:', 'D;', 'DX', 'D=']
self.emoticonsSurprise = [':-((', ':((', ':c', ':-c', ';<', ':-<', ':[', ':-[', ':-||', ":'(", ":'-(", ':0', ':-0', ':o', ':-o', ':-0']
self.emoticonsPleasant = [':3']
self.emoticonsHappy = [':)', ';) ', ':]', '];', ':P', ';P', ':D', ';D', ':->', ':-)', ';-)', ':^)', ';^)', ':-D', ':->', '=)', '=]', '<3']
self.emoticonsExcitement = ['o_o', 'O_o', 'o.O', 'O.o', 'o_0', '0_o']
```

If we follow Rule-1 blindly!



Neelita @neeldreamz · Apr 1, 2015

The first step towards Dream... [#FirstWorkingDay](#) [#Nervous](#) [#Excited](#)
[#Happy](#) [#Determined](#)



- Which of the tags we apply?

We need a Rule 3!!



	word	emotion	association
0	aback	anger	0
1	aback	anticipation	0
2	aback	disgust	0
3	aback	fear	0
4	aback	joy	0
5	aback	negative	0
6	aback	positive	0
7	aback	sadness	0
8	aback	surprise	0
9	aback	trust	0
10	abacus	anger	0
11	abacus	anticipation	0

NRC lexicon- EmoLex

- Use of Lexicon
 - List of words and their associations to eight emotions
 - Anger, fear, anticipation, trust, surprise, sadness, joy and disgust
 - Term, affect category, associationFlag (0 or 1)
- Tweets with hashtags #happy, #anger
 - All emoticons are retained in the data collection process and validated
 - Assigning weights to each emotion class for a tweet

Rule 3

- Extract the text
 - Look at the emotion association of terms
 - Assign a score for each term
 - Sum up the score for all terms
- Assign a score based on lexicon
- We can also use the NRC hastag lexicon!

NRC hashtag Lexicon

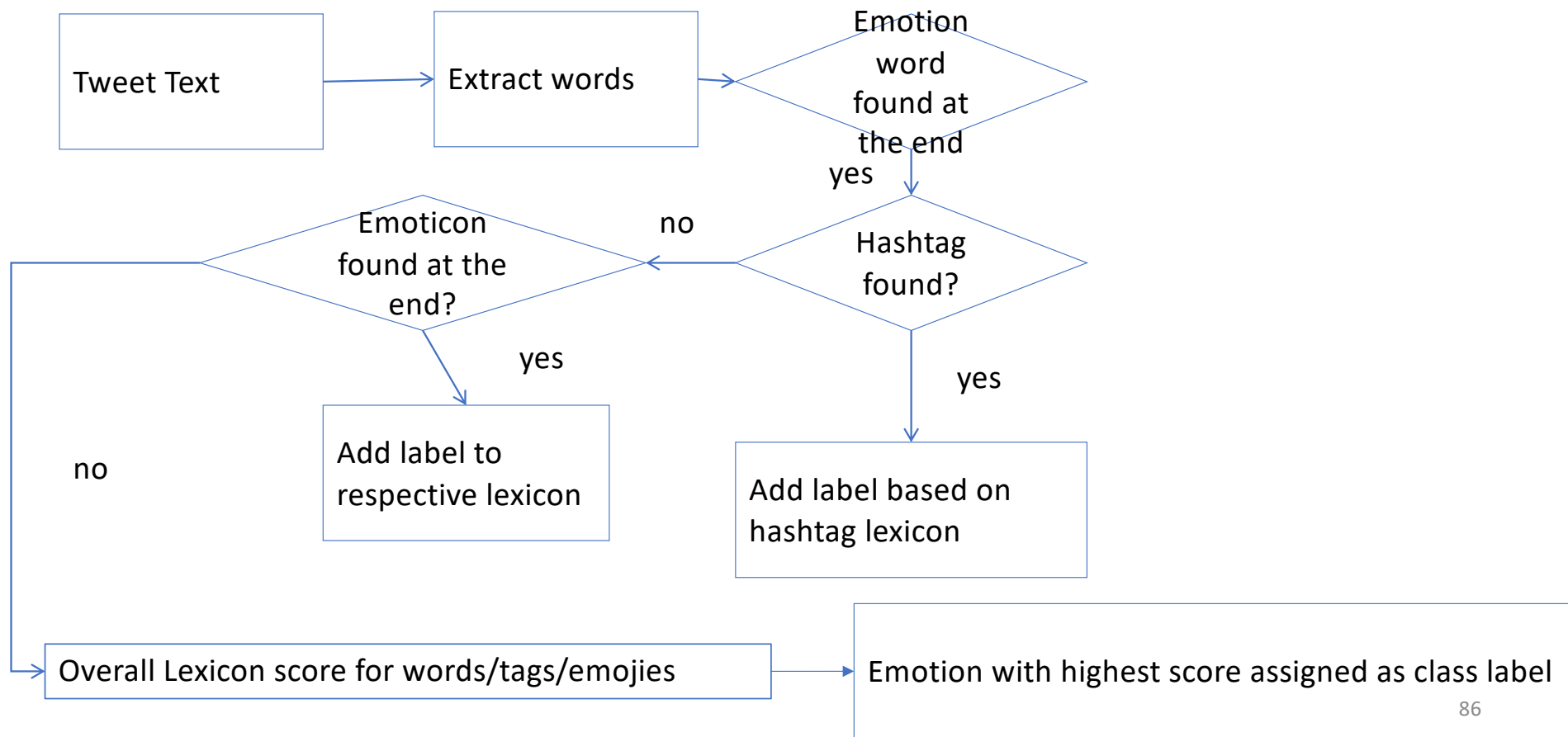
		Npos	Nneg
• .#great	8.988	6213	0
• .#classy	8.841	5364	0
• .#wonderful	8.735	4823	0
• .#perfection	8.607	4246	0
• .#perfect	8.521	3895	0
• .#excellent	8.385	3400	0
• .#positive	8.351	3285	0
• .#heavenly	8.319	3183	0
• .#fabulous	8.102	2561	0
• .#amazing	8.024	2368	0
• .#ideal	7.986	2280	0
• .#nice	7.984	2276	0
• !#amazing	7.911	2115	0
• !#positive	7.810	1912	0
• .#superb	7.663	1651	0
• !#ideal	7.603	1554	0
• !#perfect	7.568	1501	0
• ... #great	7.539	1458	0

- <score> is a real-valued sentiment score:
score = $\text{PMI}(w, \text{pos}) - \text{PMI}(w, \text{neg})$, where PMI stands for Pointwise Mutual Information between a term w and the positive/negative class;
- <Npos> is the number of times the term appears in the positive class, i.e., in tweets with positive hashtags;
- <Nneg> is the number of times the term appears in the negative class, i.e., in tweets with negative hashtags.

Compute overall score

- Hashtag at the end
 - Assign the label
- Emoticon at the end
 - Assign the label
- Lexicon score
 - For each feature (word/hashtag/emoticon)
 - Compute lexicon score
- Select the label correspondingly
 - Based on overall score

Emotion Labelling



Datasets

- Wang dataset!

Table 1. Dataset Statistics

Dataset	Anger	Fear	Sadness	Happiness	Total
CF	1311	8422	5147	5182	20062
TEC	1554	2811	3829	8235	16429
Wang	291060	71953	330060	385805	1078878



Deep Learning Models

Deep learning-based approaches

- Create large datasets
- Label them
- Train a classifier
 - 80% of data
- Validate the model
 - 10% of data
- Test the model
 - 10% of data

Datasets

Table 1. Dataset Statistics

Dataset	Anger	Fear	Sadness	Happiness	Total
CF	1311	8422	5147	5182	20062
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TEC dataset

id	text	emotion
145353048817012000	Thinks that @melbahughes had a great 50th birthday party :)	surprise
144279638024257000	Como una expresi�n tan simple, una sola oraci�n, puede llegara da�arte tanto.	sadness
140499585285111000	the moment when you get another follower and you cheer.	joy
145207578270507000	Be the greatest dancer of your life! practice daily positive habits. #fun #freedom #habits	joy
139502146390470000	eww.. my moms starting to make her annual rum cake for the whole ramdya/ally family. fml fml fml the smell!.....	disgust
146042696899887000	If ur heart hurts all the time for tht person something isn't right where's the	joy
145492569609084000	I feel awful, and it's way too freaking early. Now off to leadership highschool...	joy
145903955229151000	So chuffed for safc fans! Bet me dar comes in mortz from the match	joy
142717613234069000	Making art and viewing art are different at their core!	fear
144183822873927000	Soooo doooowwn!! Move on, get some sleep... Me deserve better. #forgetit #yawning	anger
145959995803058000	"We're sorry, but the clip you selected isn't available from your location. Please select another clip." NO I REFUSE.	sadness
142918097010036000	Een ware Sint "in-en uittocht" achter de rug; Winkel IN, winkel UIT, winkel IN, winkel UIT!	surprise
142186403223179000	People know they can pull you down and they dont give a care &	surprise
145703538276843000	My heart and soul @Jay_BabeBee is leaving me and I can't even see here	sadness
145947752709369000	chips and curry sauce	joy
138958921875464000	Soo if i hit youu , i garrentee i won't stopp . type to keep going till i make a bitch bleed foreal !	anger
146160423807361000	Oh and off to work till midnight -.-	joy
139119971073658000	I have a package at the post office. Can't think what could be in it. I don't remember internet shopping while drinking.	surprise
139925034377547000	@mizmers bahahahaha so many things i could say... RT @danecook: I just shit my pants. Pure 100% gravy.	joy
145965434397659000	@GC51 morning :) oh what a bad episode to come in towards the end... And only 1 more episode??	sadness
140532742201950000	large crowds of manic football fans & mascots	fear
145880837525733000	There is no concrete equation for joy. you can't explain how to get in your life.	joy
145160038745513000	I don't need present's of king's but just just to be in the presence of The King #Peace #PleasuresForEvermore	joy
140507961532493000	All my battles r surely handled properly & never by me ... it feels so good	joy
142739020919750000	Bringing my cousin & his friend to jingle jam & they get to see drake & i dontt ? #fml	joy
145781619024859000	Nobody up worth talking too so back to sleep I try to go!	joy
144278865152458000	Even if you're bad @ something, i cant knock you if you try. 1/2 the time, those ppl are the 1s that love it the most. Don't kill their	joy
138659624336359000	Someone on my high school football team is also called Moriarty.	fear
142458117123354000	I just absolutely lost my mind.. :(	anger

What models to use?

BERT (Bidirectional Encoder Representation with Transformers)

- In 2018, Google published a paper titled “Pre-training of deep bidirectional transformers for language understanding”.
- they introduced a language model called BERT
 - Bidirectional Encoder Representation with Transformers
- that achieved state-of-the-art performance in tasks like
 - Question-Answering,
 - Natural Language Inference,
 - Classification, and
 - General language understanding evaluation or (GLUE).

OpenAI GPT

- The OpenAI GPT and BERT use the Transformer architecture
 - use a concept called attention
- this enabled the architecture to take into account long-term dependencies
 - through the self-attention mechanism that inherently changed the way we model sequential data.
 - It uses an encoder-decoder architecture

Bert approach

- BERT uses two training paradigms:
 - Pre-training and Fine-tuning.
- **During pre-training**, the model is trained on a large dataset to extract patterns.
 - This is generally an unsupervised learning task where the model is trained on an unlabelled dataset like the data from a big corpus like Wikipedia.
- **During fine-tuning** the model is trained for downstream tasks
 - like Classification, Text-Generation, Language Translation, Question-Answering, and so forth.
- Essentially, you can download a pre-trained model and then Transfer-learn the model on your data.

Bidirectional Encoder Representation with Transformers - BERT

- Making the sentences into lower case.
- Creating vocabulary.
 - **Vocabulary** is a list of unique words in the document.

138658378867159000	sounds awful but alot of people are dying recently :((	sadness
--------------------	--	---------

```
sentences = re.sub("[.,!?-]", " ", text.lower()).split('n') # filter '!', ',', '?', '!'
word_list = list(set(" ".join(sentences).split()))
```

Special Token & Purpose

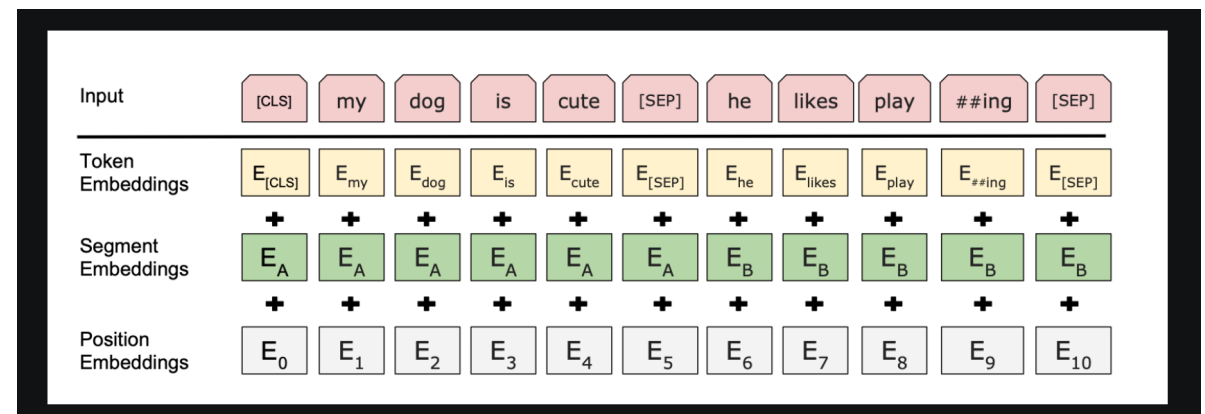
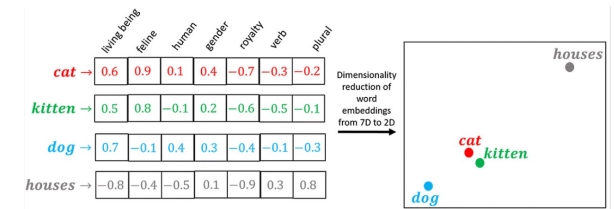
- [CLS]
 - The first token is always classification
 - [SEP]
 - Separates two sentences
 - [END]
 - End the sentence.
 - [PAD]
 - Use to truncate the sentence with equal length.
 - [MASK]
 - Use to create a mask by replacing the original word.
- `word_dict = {'[PAD]': 0, '[CLS]': 1, '[SEP]': 2, '[MASK]': 3, '[END]': 4}`
 - `for i, w in enumerate(word_list):`
 - `word_dict[w] = i + 5` `number_dict = {i: w for i, w in enumerate(word_dict)}` `vocab_size = len(word_dict)`
 - `labels_dict`
 - `{0: 'Anger', 1: 'Fear', 2: 'Sadness', 3: 'Happiness'}`

embeddings

- In the case of BERT, it creates three embeddings for
- Token,
- Segments and
- Position.

token embedding, segment embedding, and position embedding.

- “[CLS] the cat is walking [SEP] the dog is barking”.
 - [1, 5, 7, 9, 10, 2, 5, 6, 9, 11]
- A segment embedding separates
 - two sentences from each other
- A position embedding gives position to
 - each embedding in a sequence.



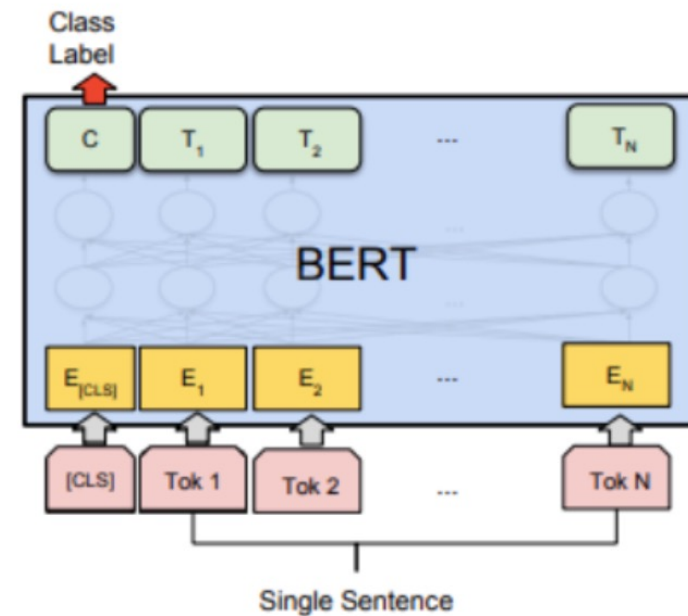
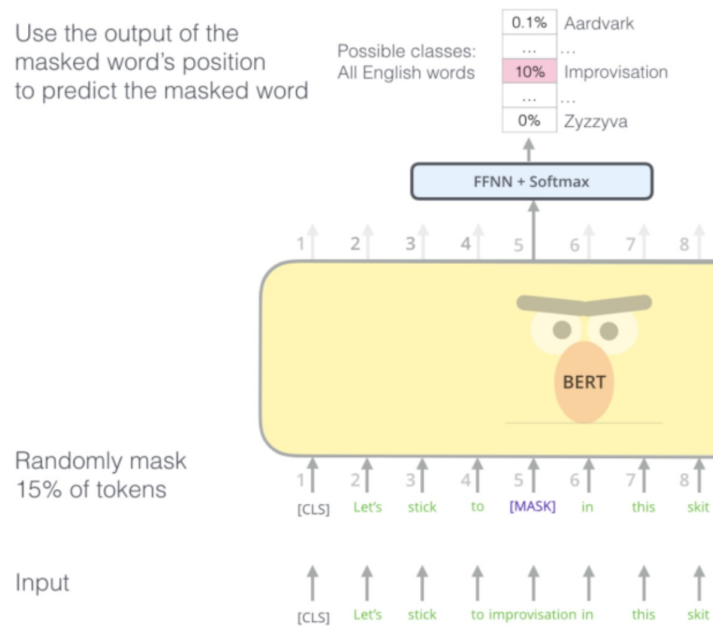
Padding

- Padding is usually done to make sure that all the sentences are of equal length.
- For instance, if we take the sentence :
- “The cat is walking. The dog is barking at the tree”
 - [CLS] The cat is walking [PAD] [PAD] [PAD].
 - [CLS] The dog is barking at the tree.”

Masking

- replace 15% of the words with [MASK] tokens, add padding.

Training & classification



With PyTorch

```
class BertClassifier(nn.Module):
    def __init__(self, model_name, num_labels, dropout_prob=0.5, bert_hidden_size=768):

        # Initializes the BertClassifier model object.

        # Parameters:
        #     model_name: {String} exact name of the Bert model to be used during training. (Usually "bert-base-uncased")
        #     num_labels: {int} Number of labels.
        #     dropout_prob : {float} The dropout probability for the classification layer.
        #     bert_hidden_size: {int} Bert hidden dimension for its last-output layer. (Usually for "bert-base-uncased" is 768)

        super().__init__()
        self.num_labels = num_labels
        self.bert = BertModel.from_pretrained(model_name)
        self.dropout = nn.Dropout(dropout_prob)
        self.ll = nn.Linear(bert_hidden_size, num_labels)
```

Forward layer

```
def forward(self, input_ids = None, attention_mask = None):
    # Processes the input data to produce the classification output and the attention weight tensor.

    # Parameters:
    #     input_data: {torch.Tensor} Input data.
    #     attention_mask: {torch.Tensor} Indices to avoid performing attention to due to padded tokens

    outputs = self.bert(input_ids, attention_mask = attention_mask)
    pooled_output = outputs[1]
    pooled_output = self.dropout(pooled_output)
    outputs = self.l1(pooled_output)
    outputs = torch.log_softmax(outputs, 1)
    return outputs
```

Using BERT to extract emotion

- Use a collection
 - Which we discussed
- Fine-tune the BERT model for emotion classification
 - 80% training data
 - Use validation dataset to choose hyper parameters (10%)
 - Test on test data set (10%)

Four emotions

- **Happiness**
- **Sadness**
- **Fear/surprise, anger**
- **Anger/disgust**

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Metrics

- True Positive
 - A +ve item correctly classified
 - False Positive (FP)
- True Negative
 - A -ve item correctly classified
 - False Negative (FN)
- Accuracy = $(TP+TN) / (TP+TN+FP+FN)$ $\longrightarrow$ Proportion of correct classification
- Precision = $TP / (TP+FP)$ $\longrightarrow$ positive items classified - what proportion is true
- Recall = $TP / (TP + FN)$ $\longrightarrow$ Positive items – what proportion of positive items correctly classified
- F1 = $2 (P * R) / (P+R)$
 - Harmonic mean

False Positive

* A -ve item classified as positive

False Negative

* A +ve item classified as Negative

How do we use it?

- Binary classification
 - Positive or negative
 - Anger vs rest
 - Happiness vs rest
- Multiclass classification
 - Anger, happiness,
- Can we create an end-to-end model?
 - Data-model –final output!

evaluation

Predicted labels

	Accuracy [STD]	Precision (WEIGHT_AVG) [STD]	Recall (WEIGHT_AVG) [STD]	F1 (WEIGHT_AVG) [STD]
TEC()	0.649 [0.010]	0.650 [0.010]	0.649 [0.010]	0.649 [0.010]

	Precision [STD]	Recall [STD]	F1 [STD]
Anger	0.622 [0.022]	0.640 [0.022]	0.631 [0.019]
Fear	0.674 [0.017]	0.677 [0.025]	0.675 [0.016]
Sadness	0.602 [0.022]	0.598 [0.024]	0.599 [0.014]
Happiness	0.703 [0.018]	0.681 [0.017]	0.692 [0.013]

How do we improve this model?

Classification results

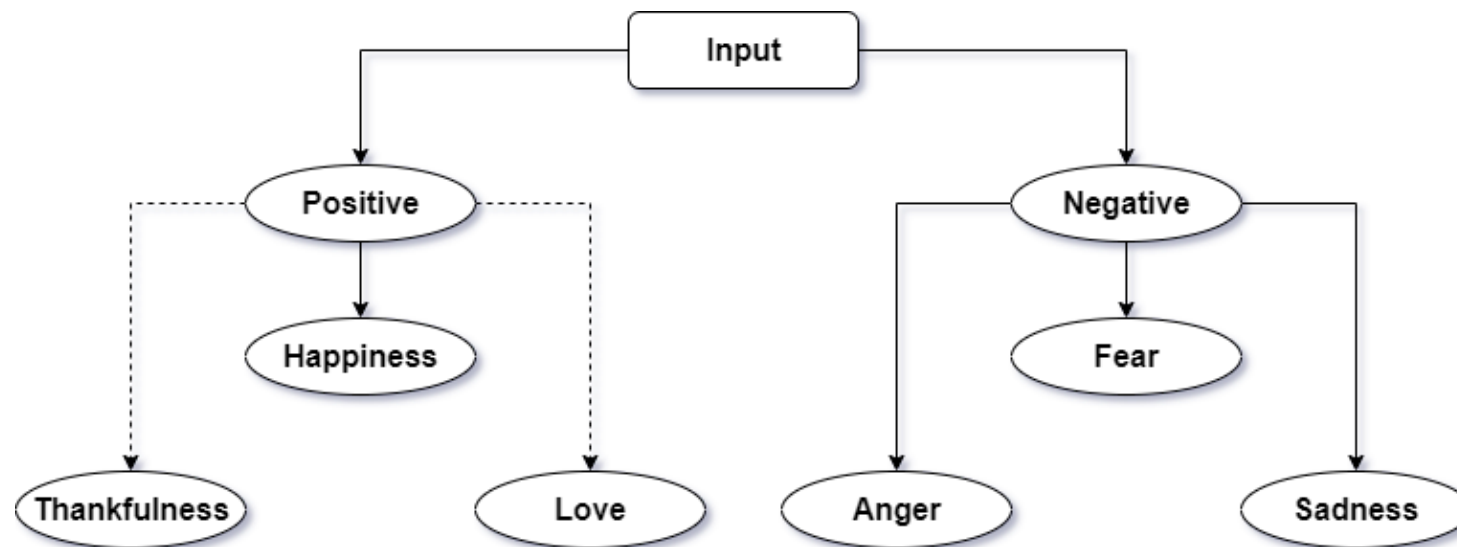
- Why do we get poor results?
 - Misclassification of emotion
 - E.g., Anger as fear!
- For sentiment classification
 - 80% or more
- If we first classify as
 - Positive or negative
 - Maybe the classification
 - Errors reduce!

Predicted labels

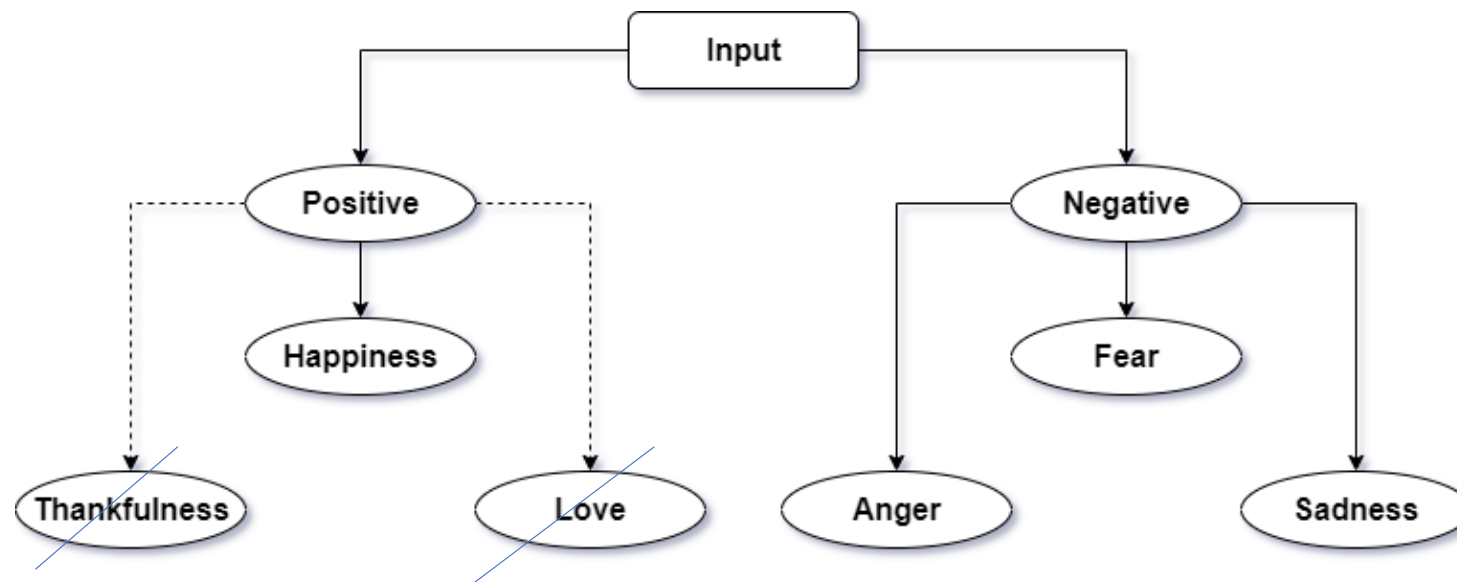
	Accuracy [STD]	Precision (WEIGHT_AVG) [STD]	Recall (WEIGHT_AVG) [STD]	F1 (WEIGHT_AVG) [STD]
TEC	0.649 [0.010]	0.650 [0.010]	0.649 [0.010]	0.649 [0.010]

	Precision [STD]	Recall [STD]	F1 [STD]
Anger	0.622 [0.022]	0.640 [0.022]	0.631 [0.019]
Fear	0.674 [0.017]	0.677 [0.025]	0.675 [0.016]
Sadness	0.602 [0.022]	0.598 [0.024]	0.599 [0.014]
Happiness	0.703 [0.018]	0.681 [0.017]	0.692 [0.013]

Hierarchical Approach



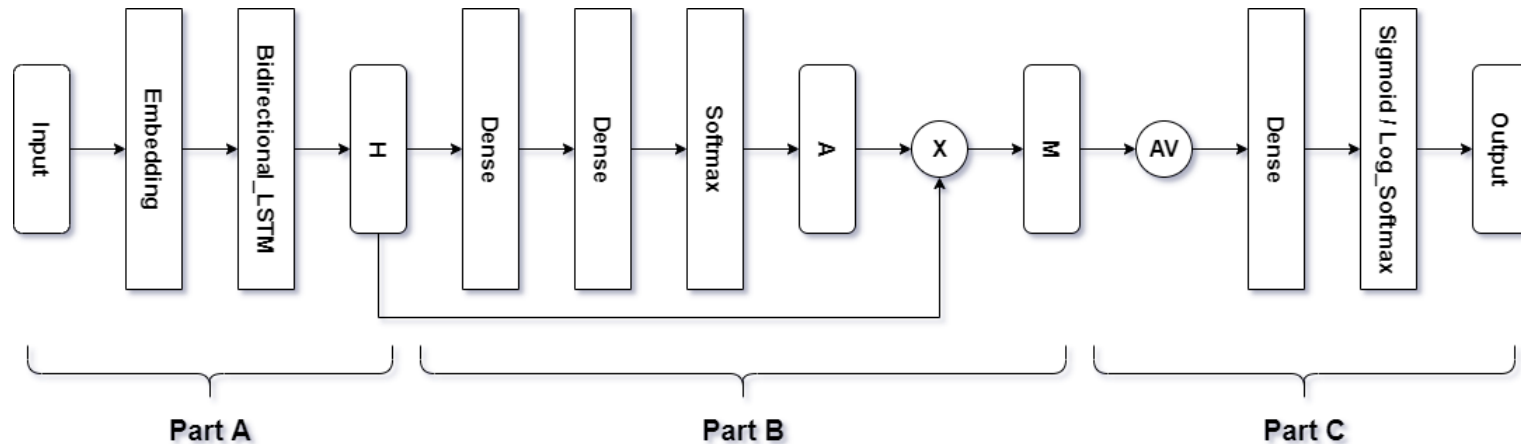
Hierarchical Approach



Deep learning models

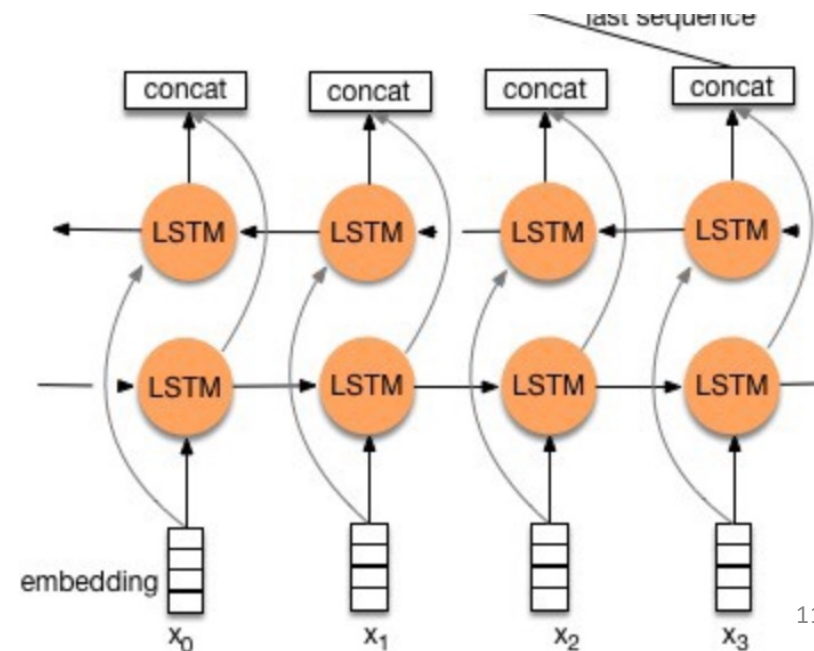
- LSTM/BiLSTM
 - Long short-term memory (LSTM) is an artificial recurrent neural network (RNN) architecture used in the field of deep learning.
- BERT

SASE model - Self-attentive sentence embedding model



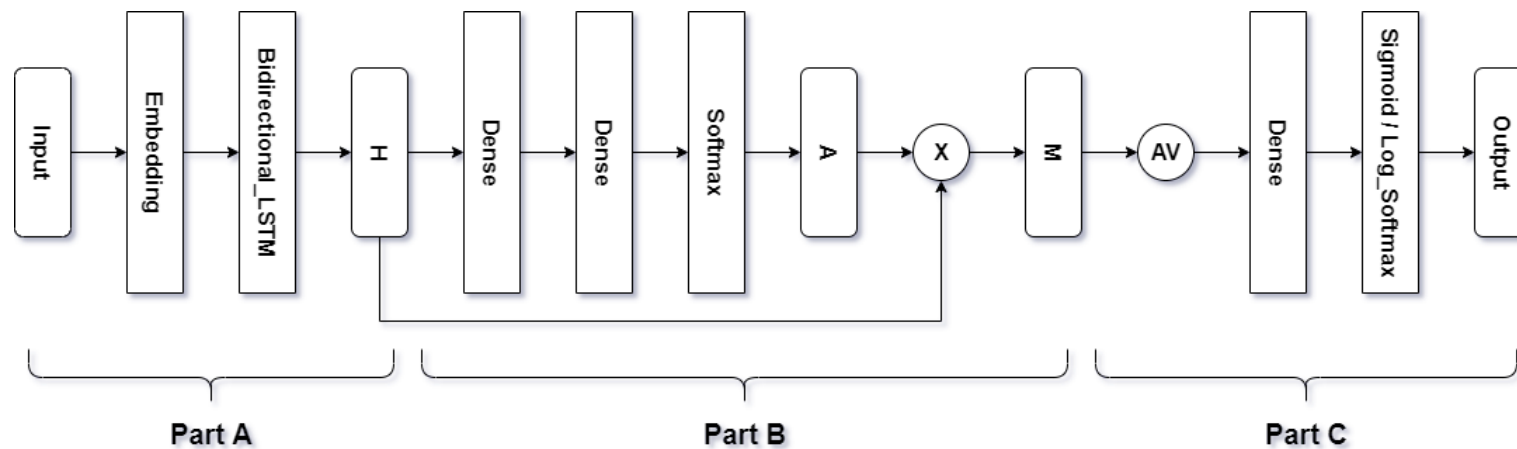
- **Part A**

- Word embedding
- biLSTM
- Long short-term memory (LSTM) is an artificial recurrent neural network (RNN) architecture used in the field of deep learning.



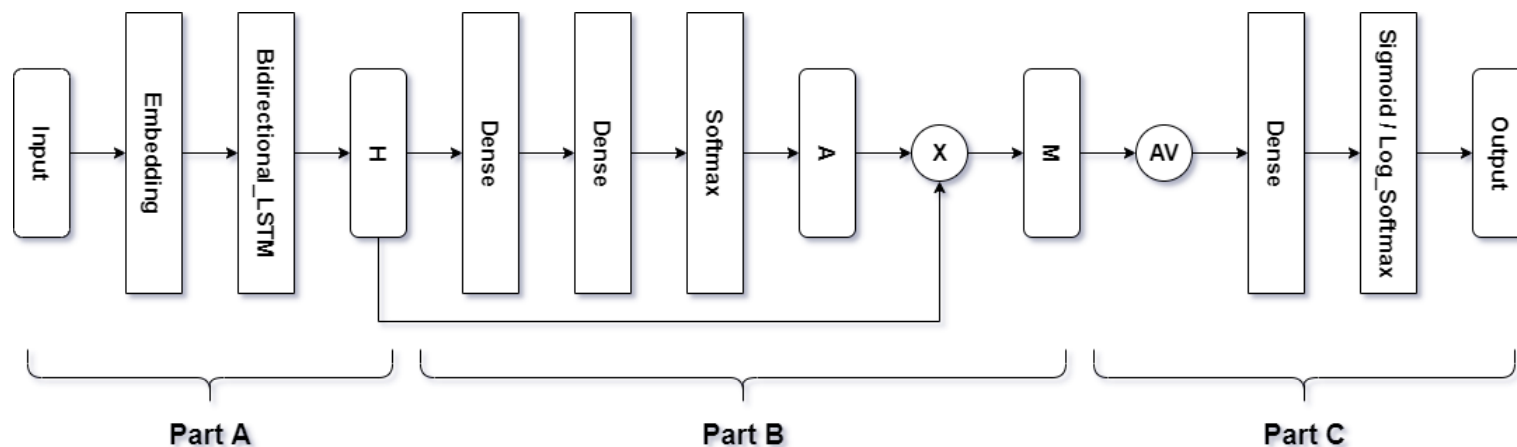
dense neural network

- layers are fully connected (dense) by the neurons in a network layer.
- Each neuron in a layer receives an input from all the neurons present in the previous layer—thus, they're densely connected

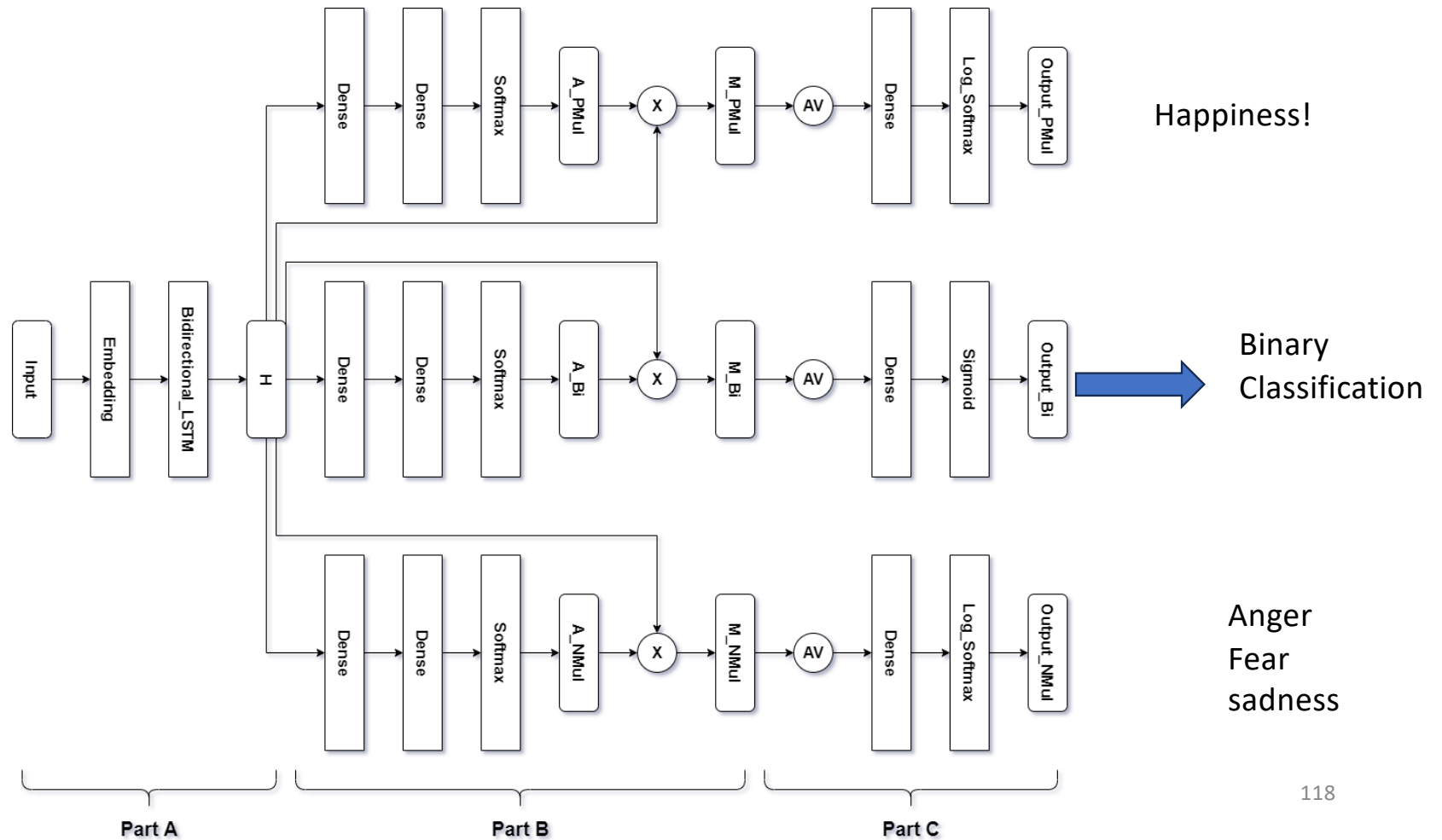


softmax function

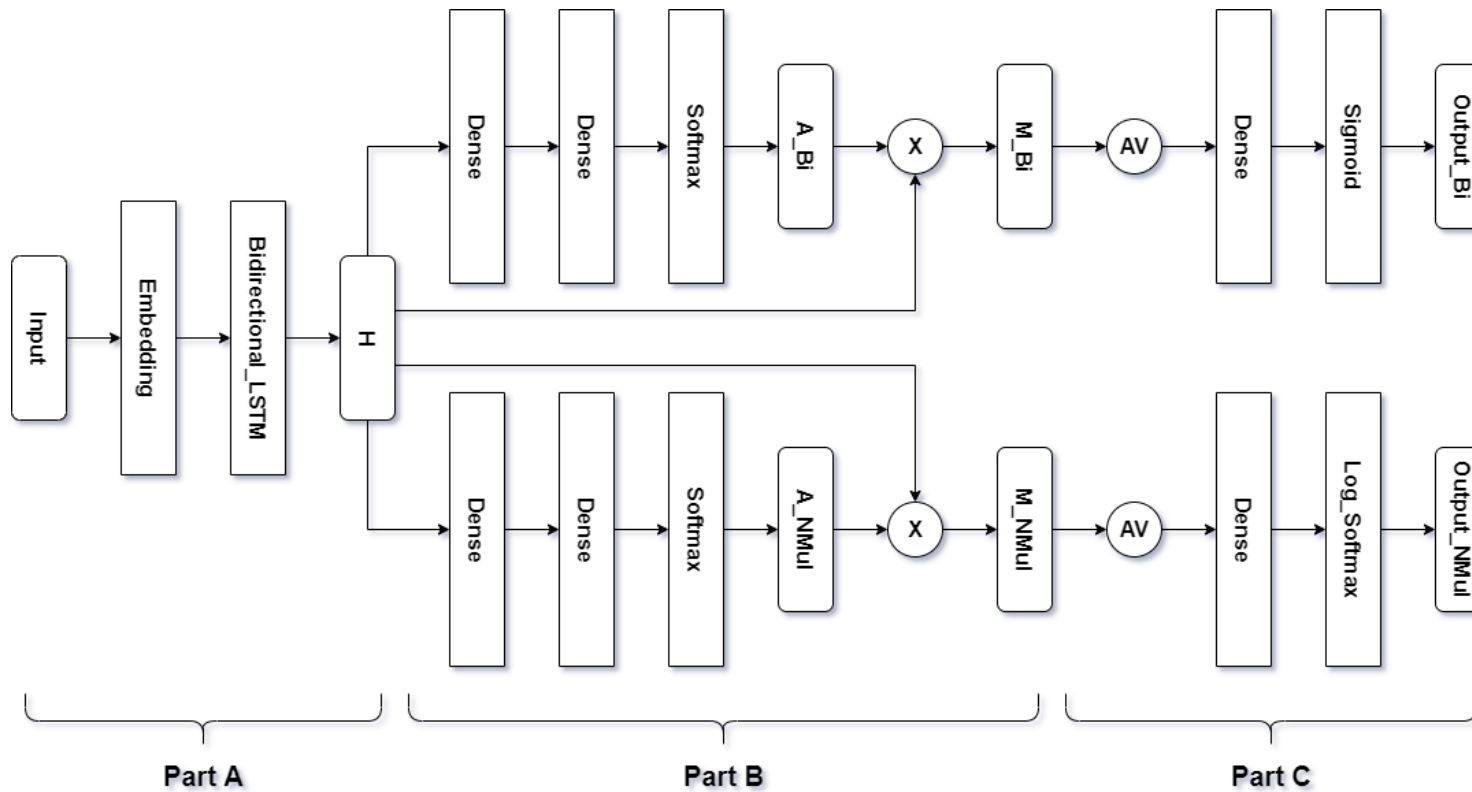
- also known as softargmax or normalized exponential function,
 - is a function that takes as input a vector of K real numbers,
 - and normalizes it into a probability distribution consisting of K probabilities proportional to the exponentials of the input numbers.



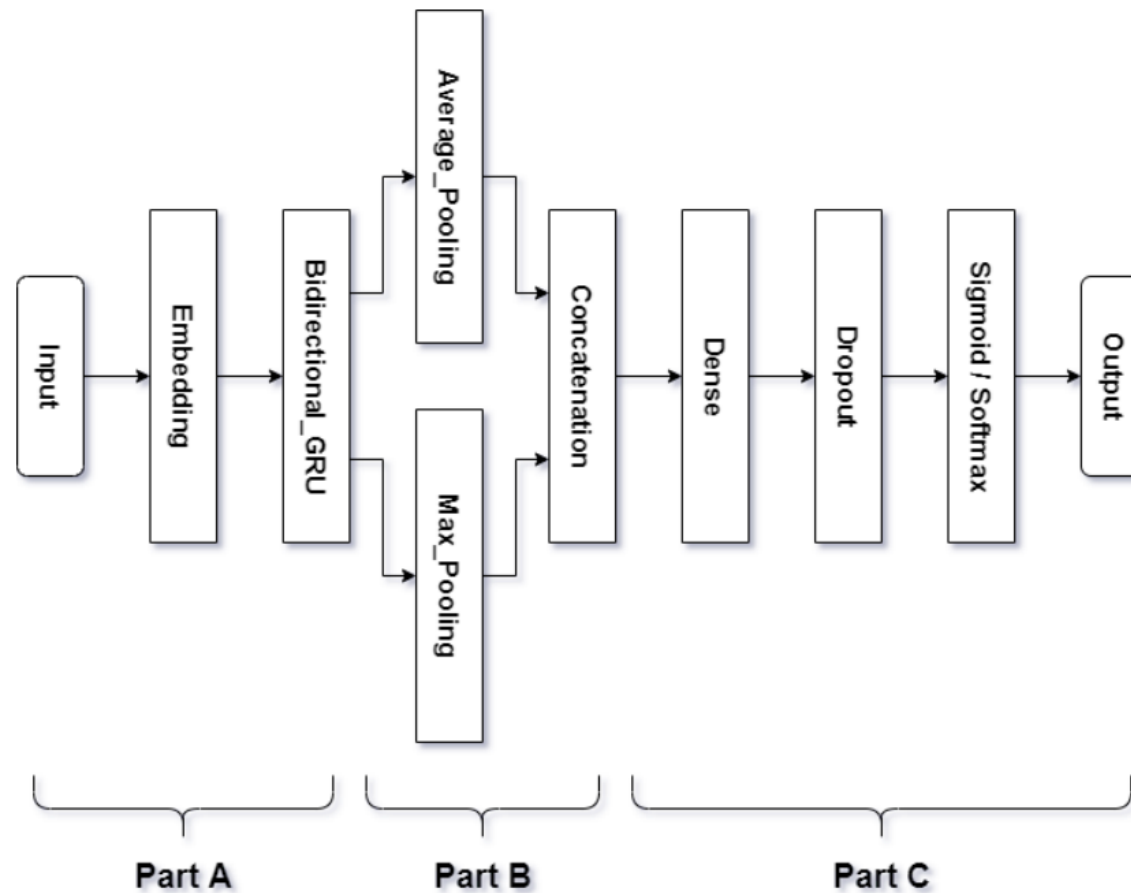
Hierarchical Approach



Hierarchical Approach



biGRU – gated recurrent unit- state of the art



Datasets

Table 1 The statistics of the datasets - percentage of samples per emotion class and total number of samples

Dataset	Anger	Fear	Sadness	Happiness	Total
CF	6.5%	42%	25.7%	25.8%	20062
TEC	9.5%	17.1%	23.3%	50.1%	16429
Wang	27.0%	6.7%	30.6%	35.7%	1078878
Vent	10.6%	12.4%	48.2%	28.8%	2259748

75% of Accuracy for sentiment classification

Evaluation

- We will use the annotated data
- 80% - for training
- 20% for testing & validation
- Do this 10 times (cross-validation)
 - 10 fold cross-validation

How do we use it?

- Binary classification
 - Positive or negative
 - Anger vs rest
 - Happiness vs rest
- Multiclass classification
 - Anger, happiness,
- Can we create an end-to-end model

Table 2 The overall results for multiclass classification

Model	Metric	CF	TEC	Wang	Vent
Bert	Accuracy	0.535	0.649	0.649	0.675
	Precision	0.530	0.651	0.650	0.678
	Recall	0.535	0.649	0.649	0.675
	F1	0.532	0.650	0.649	0.676
Bert-H	Accuracy	0.642	0.706	0.711	0.730
	Precision	0.639	0.704	0.710	0.737
	Recall	0.642	0.706	0.711	0.730
	F1	<u>0.640</u>	<u>0.704</u>	<u>0.710</u>	<u>0.733</u>
SASE	Accuracy	0.450	0.564	0.551	0.597
	Precision	0.461	0.574	0.553	0.607
	Recall	0.450	0.564	0.551	0.597
	F1	0.453	0.564	0.551	0.596
SASE-H	Accuracy	0.559	0.632	0.618	0.662
	Precision	0.575	0.636	0.625	0.685
	Recall	0.559	0.632	0.618	0.662
	F1	0.563	0.633	0.621	0.670

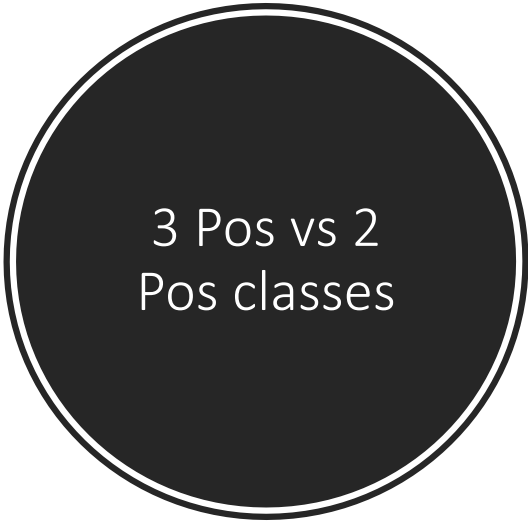
BiGRU	Accuracy	0.477	0.563	0.558	0.615
	Precision	0.482	0.574	0.562	0.619
	Recall	0.477	0.563	0.558	0.615
	F1	0.476	0.562	0.557	0.614
BiGRU-H	Accuracy	0.530	0.607	0.616	0.622
	Precision	0.554	0.629	0.632	0.655
	Recall	0.530	0.607	0.616	0.622
	F1	0.525	0.611	0.618	0.629

Classification Results

- How effective hierarchical approach?
 - Our hypothesis that an initial sentiment classification will improve the overall results?
 - Think what we did?

Looking at the results

- Sentiment classification results is 75%+
- We had only one positive class
 - Happiness
- What would have happened if we have more than one positive class?
- How do we find this out?



2POS		
Model	Metric	Wang
Bert	Accuracy	0.602
	Precision	0.603
	Recall	0.602
	F1	0.602
Bert-H	Accuracy	0.619
	Precision	0.620
	Recall	0.619
	F1	<u>0.619</u>
SASE	Accuracy	0.499
	Precision	0.501
	Recall	0.499
	F1	0.499
SASE-H	Accuracy	0.513
	Precision	0.513
	Recall	0.513
	F1	0.511
BiGRU	Accuracy	0.505
	Precision	0.509
	Recall	0.505
	F1	0.505
BiGRU-H	Accuracy	0.498
	Precision	0.502
	Recall	0.498
	F1	0.496

Table 2 The overall results for multiclass classification

Model	Metric	CF	TEC	Wang
Bert	Accuracy	0.535	0.649	0.649
	Precision	0.530	0.651	0.650
	Recall	0.535	0.649	0.649
	F1	0.532	0.650	0.649
Bert-H	Accuracy	0.642	0.706	0.711
	Precision	0.639	0.704	0.710
	Recall	0.642	0.706	0.711
	F1	<u>0.640</u>	<u>0.704</u>	<u>0.710</u>
SASE	Accuracy	0.450	0.564	0.551
	Precision	0.461	0.574	0.553
	Recall	0.450	0.564	0.551
	F1	0.453	0.564	0.551
SASE-H	Accuracy	0.559	0.632	0.618
	Precision	0.575	0.636	0.625
	Recall	0.559	0.632	0.618
	F1	0.563	0.633	0.621
BiGRU	Accuracy	0.477	0.563	0.558
	Precision	0.482	0.574	0.562
	Recall	0.477	0.563	0.558
	F1	0.476	0.562	0.557
BiGRU-H	Accuracy	0.530	0.607	0.616
	Precision	0.554	0.629	0.632
	Recall	0.530	0.607	0.616
	F1	0.525	0.611	0.618

3POS			
Model	Metric	Wang	Vent
Bert	Accuracy	0.588	0.579
	Precision	0.589	0.583
	Recall	0.588	0.579
	F1	0.588	0.579
Bert-H	Accuracy	<u>0.599</u>	<u>0.580</u>
	Precision	<u>0.598</u>	0.583
	Recall	0.599	0.580
	F1	<u>0.599</u>	<u>0.580</u>
SASE	Accuracy	0.494	0.483
	Precision	0.497	0.493
	Recall	0.494	0.483
	F1	0.494	0.482
SASE-H	Accuracy	0.491	0.506
	Precision	0.489	0.513
	Recall	0.491	0.506
	F1	0.488	0.506
BiGRU	Accuracy	0.496	0.512
	Precision	0.499	0.519
	Recall	0.496	0.512
	F1	0.495	0.512
BiGRU-H	Accuracy	0.475	0.491
	Precision	0.478	0.501
	Recall	0.475	0.491
	F1	0.475	0.492

Summary

- As the number of classes increased
 - Performance decreased
- This maybe due to cross-talk between positive emotion classes
- It is in fact
 - People rarely describe feeling a specific positive emotion without also claiming to feel other positive emotions
- This means we need to develop datasets which are more accurate ground truth!